

The effect of lifestyle modification and cardiovascular risk factor reduction on erectile dysfunction: a systematic review and meta-analysis.

BACKGROUND: Erectile dysfunction (ED) shares similar modifiable risks factors with coronary artery disease (CAD). Lifestyle modification that targets CAD risk factors may also lead to improvement in ED. We conducted a systematic review and meta-analysis of randomized controlled trials evaluating the effect of lifestyle interventions and pharmacotherapy for cardiovascular (CV) risk factors on the severity of ED. **METHODS:** A comprehensive search of multiple electronic databases through August 2010 was conducted using predefined criteria. We included randomized controlled clinical trials with follow-up of at least 6 weeks of lifestyle modification intervention or pharmacotherapy for CV risk factor reduction. Studies were selected by 2 independent reviewers. The main outcome measure of the study is the weighted mean differences in the International Index of Erectile Dysfunction (IIEF-5) score with 95% confidence intervals (CIs) using a random effects model. **RESULTS:** A total of 740 participants from 6 clinical trials in 4 countries were identified. Lifestyle modifications and pharmacotherapy for CV risk factors were associated with statistically significant improvement in sexual function (IIEF-5 score): weighted mean difference, 2.66 (95% CI, 1.86-3.47). If the trials with statin intervention (n = 143) are excluded, the remaining 4 trials of lifestyle modification interventions (n = 597) demonstrate statistically significant improvement in sexual function: weighted mean difference, 2.40 (95% CI, 1.19-3.61). **CONCLUSION:** The results of our study further strengthen the evidence that lifestyle modification and pharmacotherapy for CV risk factors are effective in improving sexual function in men with ED.